LUMINA ENERGY · HELIOS PLATFORM

Disaster Recovery Plan

CLASSIFICATION: INTERNAL — RESTRICTED

Owner: Marcus Webb (SRE Lead) + Rosa Lindqvist (Senior Platform Engineer) | Last DR Test: 2024-09-18 | Next Scheduled: 2025-03-12 | Last Updated: 2024-10-22

■ Rosa, 2024-10-22: Updated RTO/RPO targets following the September DR test — we didn't meet the previous 2-hour RTO. Revised to 4 hours based on actual results. Aurora cross-region replication lag was 12 min vs. 5-min design target. INFRA-2847 opened.

■ Marcus, 2024-09-19: Kafka cross-region replay took 3.5 hours, not 90 minutes. IoT device re-registration gap is unresolved. We have contractual SLAs with NW Grid UK and ENGIE referencing a 4-hour RTO.

1. Recovery Objectives

Targets — Revised After September 2024 Test

METRIC	TARGET	SEP 2024 ACTUAL	STATUS
RTO — Recovery Time Objective	4 hours	4h 22min	■ Near Miss
RPO — Recovery Point Objective	15 minutes	12 minutes	✓ Met
Grid telemetry restoration	2 hours	1h 48min	✓ Met
Customer portal availability	3 hours	3h 05min	■ Near Miss
AI forecasting restoration	6 hours	5h 51min	✓ Met
Dispatch service restoration	4 hours	4h 17min	■ Near Miss

Priority Order for Service Restoration

PRIORITY	WINDOW	SERVICES
Priority 1 Life Safety / Grid Safety	0–90 min	Grid telemetry ingestion Outage detection service Emergency dispatch API Database (read-only minimum)
Priority 2 Core Operations	90 min – 3 hrs	Full dispatch service Customer notifications Grid monitoring portal
Priority 3 Business Functions	3–6 hrs	Customer portal AI forecasting engine GIS mapping service Analytics platform

Priority 4 Non-Critical	6+ hrs	Reporting Historical data queries Admin portal

2. Architecture Overview

Helios runs primarily in eu-west-1 (Ireland) with a warm standby in eu-central-1 (Frankfurt). The multi-region design was established in ADR-007 (Architecture Decision Records).

COMPONENT	REPLICATION METHOD	TYPICAL LAG	MAX OBSERVED
Aurora PostgreSQL	Aurora Global Database	2–5 min	12 min (Sep 2024 test)
TimescaleDB (grid telemetry)	Aurora Global Database	2–5 min	12 min
Redis	No replication — regenerated on failover	N/A	N/A
Kafka	MirrorMaker 2	30–90 sec	8 min (under load)
S3 (ML models, reports)	Cross-region replication	< 15 min	22 min
ECR (container images)	Cross-region replication	< 30 min	N/A

3. Known Gaps and Risks

ID	DESCRIPTION	RISK	OWNER	TARGET
IoT Gap	IoT devices cannot auto re-register in secondary region during failover (42M devices)	HIGH	Lars Eriksson	Q1 2025
Kafka MM2 Consumer Offset Drift	Consumer group offsets not reliably sync'd; may reprocess on failover	MED	David Okafor	Accepted
Aurora Lag Under Load	Replication lag reached 12 min in test vs. 5-min design target	MED	Rosa Lindqvist	INFRA-2847
No Chaos Engineering	Only 2 manual DR tests in platform history; no automated chaos tooling	LOW	Marcus Webb	2025 planning

4. September 2024 Full DR Test Results

STEP	TARGET	ACTUAL	RESULT
DR decision made	T+15 min	T+12 min	✓
Aurora promotion initiated	T+20 min	T+22 min	✓
Aurora promotion complete	T+35 min	T+36 min	✓
DNS failover complete	T+40 min	T+43 min	✓
Priority 1 services live	T+90 min	T+1h 48min	✓
Kafka consumers caught up	T+3 hrs	T+6h 30min	✗ **Blocker**
Customer portal live	T+3 hrs	T+3h 05min	■

AI forecasting live	T+6 hrs	T+5h 51min	✓
Full service restoration	T+4 hrs	T+7h 22min	✗ **Blocker**

Key finding: Kafka MirrorMaker lag was ~8 hours at time of test (MirrorMaker had been unhealthy for 2 weeks, undetected). INFRA-2848 (MirrorMaker health monitoring) completed 2024-10-03. INFRA-2847 (Aurora lag) and INFRA-2901 (IoT cross-region) remain open.